Carrying out projections onto specially structured sets

- To do projections onto sets other than a box, we need some special tools.

- Consider the problem

$$\begin{aligned} \min \quad & f(x) \\ \text{st} \quad & g(x) = b \\ & x \in X \end{aligned} \qquad (\ddagger)$$

where $f: \mathbb{R}^n \to \mathbb{R}$, $g: \mathbb{R}^n \to \mathbb{R}$

- For fixed $\lambda \in \mathbb{R}$, consider the problem

$$\begin{aligned} \min \quad & f(x) + \lambda(g(x) - b) \\ \text{st} \quad & x \in X \end{aligned} \qquad (*)$$

- Let $x^*(\lambda)$ be a unique optimal solution to problem $(*)$.

Find value of λ^* such that $g(x^*(\lambda^*)) = b$

- Then $x^*(\lambda^*)$ is an optimal solution to problem $(\ddagger)$

Pf: • Let $\hat{x}$ be an optimal solution $(\ddagger)$. So, $g(\hat{x}) = b$, $\hat{x} \in X$

Also, $x^*(\lambda^*)$ satisfies $g(x^*(\lambda^*)) = b$ $\quad x^*(\lambda^*) \in X$

So we get $\quad f(\hat{x}) \leq f(x^*(\lambda^*))$

• $x^*(\lambda^*)$ is an optimal solution to $(*)$ with $\lambda = \lambda^*$, so $x^*(\lambda^*) \in X$, $g(x^*(\lambda^*)) = b$

Also we have $\quad \hat{x} \in X$, $\quad g(\hat{x}) = b$

Because $x^*(\lambda^*)$ is an opt. solution to $(*)$, we get

$$f(x^*(\lambda^*)) + \lambda^* \cancelto{0}{(g(x^*(\lambda^*)) - b)} \leq f(\hat{x}) + \lambda^* \cancelto{0}{(g(\hat{x}) - b)}$$

$$f(x^*(\lambda^*)) \leq f(\hat{x})$$

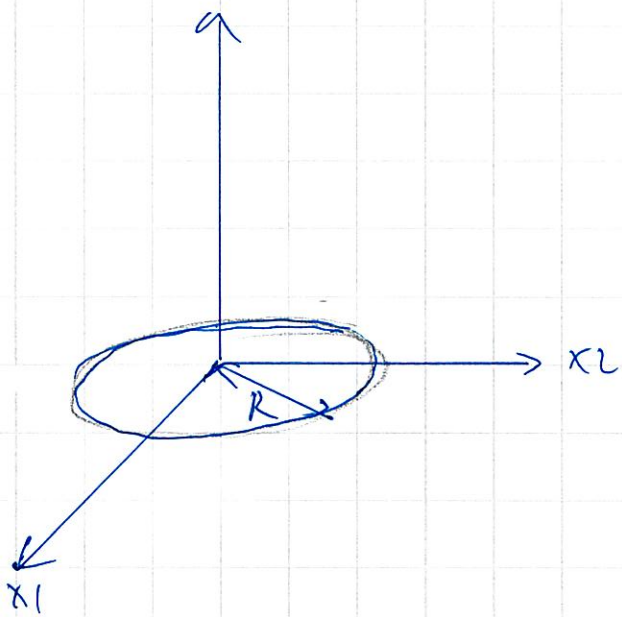
• Thus, we have $\quad f(x^*(\lambda^*)) = f(\hat{x})$. Also we have $x^*(\lambda^*) \in X$, $g(x^*(\lambda^*)) = b$.

So $x^*(\lambda^*)$ is a feasible solution to $(\ddagger)$ and gives an objective value equal to the opt. obj. value of this problem

Thus, $x^*(\lambda^*)$ must be optimal to $(\ddagger)$ $\qquad$ done!

Projections on to the Euclidean Ball

- Consider the case $X = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^{n} x_i^2 \leq R^2 \right\}$



- We want to find $\Pi_X(\hat{y})$, where $\hat{y}$ is a fixed point.
- If $\hat{y} \in X$, then $\Pi_X(\hat{y}) = \hat{y}$, so consider the case $\hat{y} \notin X$, which means $\sum_{i=1}^{n} (\hat{y}_i)^2 > R^2$

- We want to solve $\min \sum_{i=1}^{n} (x_i - \hat{y}_i)^2$ (*)
  st $\sum_{i=1}^{n} (x_i)^2 = R^2$

- For fixed λ, consider the problem $\min_x \left\{ \sum_{i=1}^{n} (x_i - \hat{y}_i)^2 + \lambda \left(\sum_{i=1}^{n} x_i^2 - R^2 \right) \right\}$

- The last problem is equivalent to:

$$\sum_{i=1}^{n} \min_x \left\{ (x_i - \hat{y}_i)^2 + \lambda x_i^2 \right\} - \lambda R^2$$

The opt. solution to each minimization problem above is given by

$$\cancel{2}(x_i - \hat{y}_i) + \cancel{2}\lambda x_i = 0 \Rightarrow x_i = \frac{1}{1+\lambda} \hat{y}_i \Rightarrow x_i^*(\lambda) = \frac{1}{1+\lambda} \hat{y}_i$$

- Find λ^* such that $g(x^*(\lambda^*)) = 0$, which means $\sum_{i=1}^{n} (x_i^*(\lambda^*))^2 = R^2$

So we want $\sum_{i=1}^{n} \frac{1}{(1+\lambda^*)^2} (\hat{y}_i)^2 = R^2 \Rightarrow \frac{1}{(1+\lambda^*)^2} \|\hat{y}\|^2 = R^2 \Rightarrow \frac{1}{1+\lambda^*} = \frac{R}{\|\hat{y}\|}$

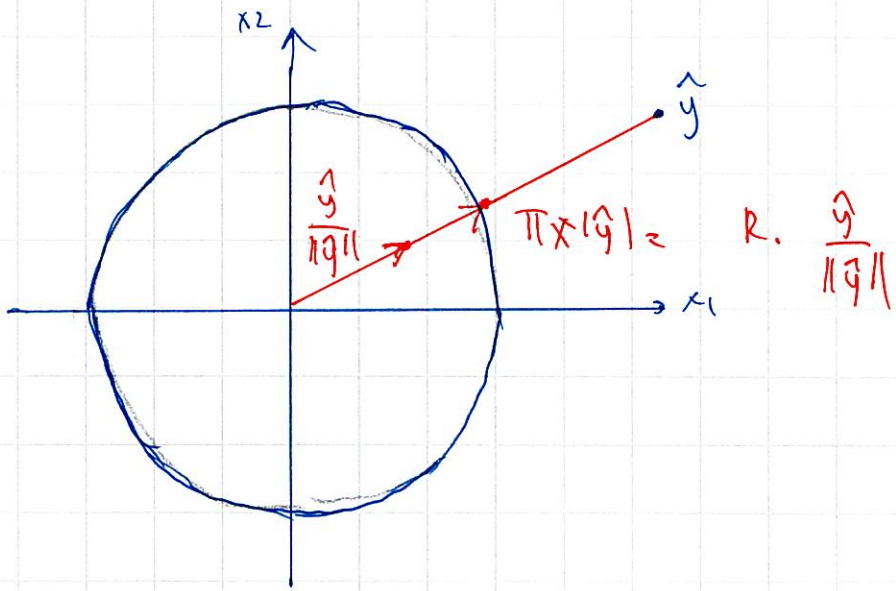
- Therefore $x_i^*(\lambda^*) = \dfrac{1}{1+\lambda^*}\,\hat{y}_i = \dfrac{R}{\|\hat{y}\|}\cdot\hat{y}_i$

- In this case, the opt. solution to $(\ddagger)$ is given by x^*, where

$$x_i^* = \frac{R}{\|\hat{y}\|}\cdot\hat{y}_i$$

- Thus, the projection of $\hat{y}$ onto Euclidean ball is given by π

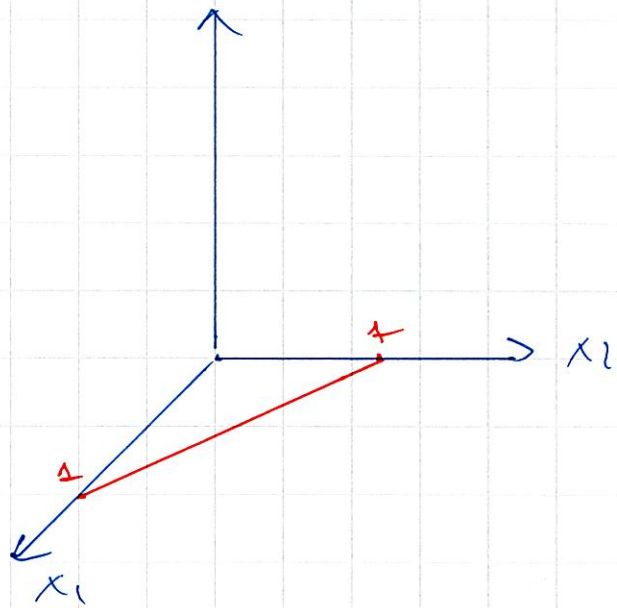
$$\pi_i = \frac{R}{\|\hat{y}\|}\cdot\hat{y}_i \qquad \left(\pi = \frac{R}{\|\hat{y}\|}\cdot\hat{y}\right)$$



Projections onto Euclidean ball is useful for ridge regression.

Projection onto a Probability Simplex

Consider the case $X = \{ x \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0 \ \forall i = 1, \dots, n \}$



We want to find $\Pi_X(\hat{y})$, where $\hat{y}$ is a fixed point.

- We want to solve

$$\min \sum_{i=1}^{n} (x_i - \hat{y}_i)^2$$

$$\text{st} \quad \sum_{i=1}^{n} x_i = 1$$

$$x_i \geq 0 \quad \forall i = 1, \dots, n$$

- For fixed λ, consider the problem

$$\min_{\text{st}} \left\{ \sum_{i=1}^{n} (x_i - \hat{y}_i)^2 + \lambda \left(\sum_{i=1}^{n} x_i - 1 \right) \right\}$$

$$x_i \geq 0 \quad \forall i = 1, \dots, n$$

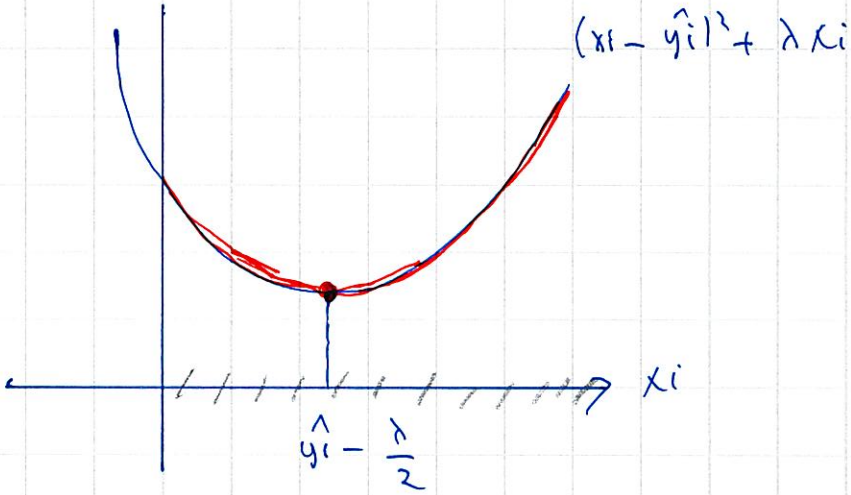
- The last problem is equivalent to

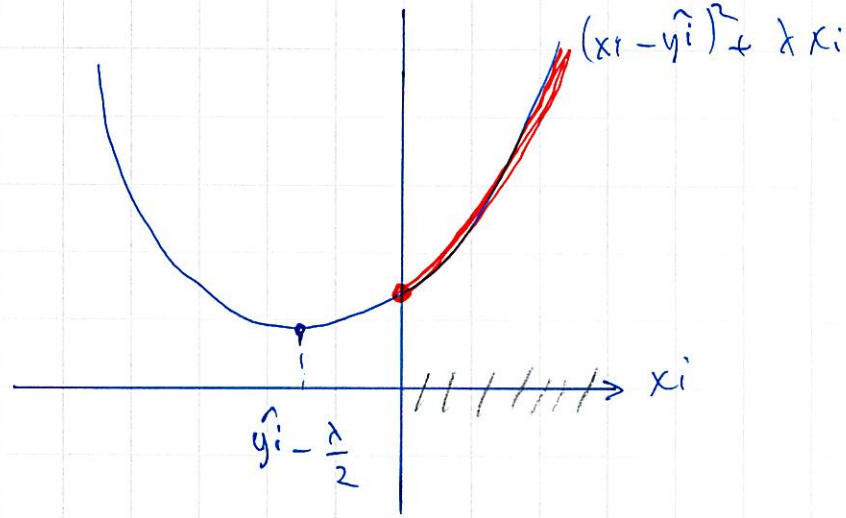
$$\sum_{i=1}^{n} \min_{x_i \geq 0} \left\{ (x_i - \hat{y}_i)^2 + \lambda x_i \right\} - \lambda$$

- To solve each minimization problem above,

$$2(x_i - \hat{y}_i) + \lambda = 0 \Rightarrow x_i = \hat{y}_i - \frac{\lambda}{2}$$

Thus, we get $x_i^*(\lambda) = \left[\hat{y}_i - \frac{\lambda}{2} \right]^+$


$(x_i - \hat{y}_i)^2 + \lambda x_i$
x_i
$\hat{y}_i - \frac{\lambda}{2}$


$(x_i - \hat{y}_i)^2 + \lambda x_i$
x_i
$\hat{y}_i - \frac{\lambda}{2}$

- Find λ^* such that $g(x^*(\lambda^*)) = b$, which means $\sum_{i=1}^{n} x_i^*(\lambda^*) = 1$

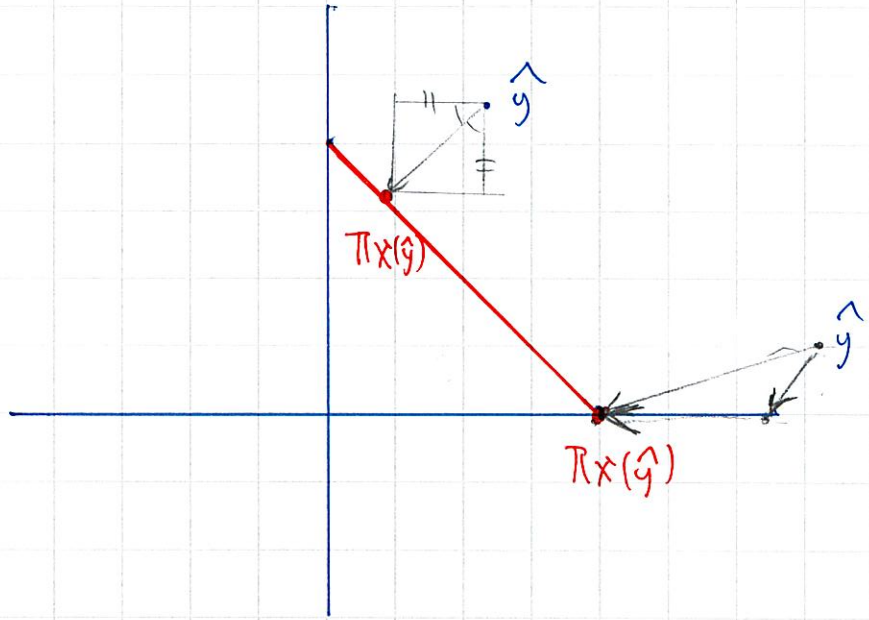
So we want $\sum_{i=1}^{n} \left[\hat{y_i} - \frac{\lambda^*}{2} \right]^+ = 1$

- Find λ^* such that $\sum_{i=1}^{n} \left[\hat{y_i} - \frac{\lambda^*}{2} \right]^+ = 1$ (which is always possible)

- then $x_i^*(\lambda^*) = \left[\hat{y_i} - \frac{\lambda^*}{2} \right]^+$

- Thus, to project $\hat{y}$ on to the probability simplex, find λ^* such that $\sum_{i=1}^{n} \left[\hat{y_i} - \frac{\lambda^*}{2} \right]^+ = 1$, then the projection π is given by

$$\pi_i = \left[\hat{y_i} - \frac{\lambda^*}{2} \right]^+$$


$\hat{y}$
$\Pi_X(\hat{y})$
$\hat{y}$
$\Pi_X(\hat{y})$